

FORCE MULTIPLIER

Key upgrades of Rafale F4. The IAF operates the F3R version

Radar: The Rbe2 AESA Radar is upgraded with new algorithms, including the Ground Mobile Target Indication and Tracking (GMTI/T) for better sensor fusion

Avionics: Upgraded SPECTRA Electronic Warfare (EW) suite, with improved threat detection, better electronic jamming capability, stronger survivability against missiles

↘ Enhanced data links; improved satellite communication; real-time battlefield data-sharing (network-centric warfare); better integration with AWACS, drones, ground systems

New weapons: MICA NG next-gen air-to-air missile; future compatibility with upgraded precision-guided munitions



"INDIA HAS A QUALITATIVELY IMPRESSIVE AIR FORCE, BUT IT FACES A QUANTITATIVE DISADVANTAGE AGAINST CHINESE AIRPOWER, AND THAT DISADVANTAGE IS GROWING. RAFALE HELPS TO STAVE OFF THE TREND"

CHRISTOPHER CLARY
Non-resident fellow, Stimson Center

A PROGRAMME RERUN

This latest move to acquire Rafales through the MRFA programme is a rerun of the Medium Multi-role Combat Aircraft (MMRCA) tender launched in 2007. After years of trials, the Rafale was selected in 2012 as the preferred fighter based on superior performance and lifecycle sustainability. Negotiations commenced for the direct purchase of 18 aircraft in fly-away condition, alongside the licensed manufacture of the remaining 108 fighters by state-owned Hindustan Aeronautics Limited (HAL) through technology transfer. In 2015, the new NDA government scrapped the MMRCA deal and opted to buy 36 Rafales directly through an inter-governmental agreement signed in September 2016, with deliveries completed by the end of 2022. The current deal does not have HAL as a production partner as it is committed to multiple ongoing projects.

Air Marshal Anil Chopra (retd), a member of the technical evaluation committee for the MMRCA deal, says that the IAF was satisfied with the Rafale's performance during Operation Sindoor, especially in the way it jammed all Pakistan Air Force frequencies. He also points out that the French are now offering the latest Rafale F4 version and subsequently F5, featuring gallium nitride–based electronic warfare (EW) systems, along with the latest active electronically scanned array (AESA) radar. The IAF operates the Rafale F3R version.

The assembly line for the Indian production of 96 Rafales is expected to be set up at the Dassault Reliance Aerospace Limited (DRAL) facility in Nagpur, a subsidiary of Dassault Aviation after it acquired a majority stake in September last year. The programme will involve multiple Indian companies, including the Tata Group, Mahindra and Dynamatic Technologies Ltd, along with over 30 other suppliers. Tata is already